

$$\min z = x_1 + x_2 + x_3$$

$$\text{s.t. } 2x_1 + 7.5x_2 + 3x_3 \geq 10000$$

$$20x_1 + 5x_2 + 10x_3 \geq 30000$$

$$x_1, x_2, x_3 \geq 0$$

1. 先转换为约束标准型

① 引入松弛变量 x_4, x_5

$$2x_1 + 7.5x_2 + 3x_3 - x_4 = 10000$$

$$20x_1 + 5x_2 + 10x_3 - x_5 = 30000$$

② 系数矩阵无单位阵, 引入人工变量 x_6, x_7

$$2x_1 + 7.5x_2 + 3x_3 - x_4 + x_6 = 10000$$

$$20x_1 + 5x_2 + 10x_3 - x_5 + x_7 = 30000$$

③ 用两阶段法求解

$$\max z' = -x_1 - x_2 - x_3, \quad \min W = x_6 + x_7, \quad \max w' = -x_6 - x_7$$

$$\text{s.t. } 2x_1 + 7.5x_2 + 3x_3 - x_4 + x_6 = 10000 \quad (1)$$

$$20x_1 + 5x_2 + 10x_3 - x_5 + x_7 = 30000 \quad (2)$$

$$x_1, x_2, x_3, x_4, x_5, x_6, x_7 \geq 0$$

2. 构造单纯形表, $w' = -x_6 - x_7 = 22x_1 + 12.5x_2 + 13x_3 - x_4 - x_5 - 40000$

常数项	x_1	x_2	x_3	x_4	x_5
$w \quad -40000$	22	12.5	13	-1	-1
$x_6 \quad 10000$	2	7.5	3	-1	0
$x_7 \quad 30000$	20	5	10	0	-1

选 x_1 为入基变量 (w 行最大正系数 22)

$$x_6 \text{ 行 } \frac{10000}{2} = 5000, x_7 \text{ 行 } \frac{30000}{20} = 1500$$

选 x_7 为离基变量.

$$x_1 + \frac{1}{4}x_2 + \frac{1}{2}x_3 - \frac{1}{20}x_5 + \frac{1}{20}x_7 = 1500$$

代入 (1) 得

$$7x_2 + 3x_3 - x_4 + 0.1x_5 + x_6 - 0.1x_7 = 7000$$

$$w = 22x_1 + 12.5x_2 + 13x_3 - x_4 - x_5 - 40000$$

$$\begin{aligned} & 22x_1 + 5.5x_2 + 11x_3 - 1.1x_5 + 1.1x_7 \\ &= 7x_2 + 2x_3 - x_4 + 0.1x_5 - 1.1x_7 - 7000 \end{aligned}$$

	常数项	x_2	x_3	x_4	x_5	x_7
w	-7000	7	2	-1	0.1	-1.1
x_6	7000	7	2	-1	0.1	-0.1
x_1	1500	$\frac{1}{4}$	$\frac{1}{2}$	0	$-\frac{1}{20}$	$\frac{1}{20}$

选 x_2 为入基变量

$$x_6 \text{ 行: } \frac{7000}{7} = 1000, x_7 \text{ 行 } \frac{1500}{1/4} = 6000$$

选 x_6 行为离基变量

$$x_2 + \frac{2}{7}x_3 - \frac{1}{7}x_4 + \frac{1}{70}x_5 - \frac{1}{7}x_6 - \frac{1}{70}x_7 = 1000$$

$$\text{代入 } x_1 + \frac{1}{4}x_2 + \frac{1}{2}x_3 - \frac{1}{20}x_5 + \frac{1}{20}x_7 = 1500$$

$$\Rightarrow x_1 + \frac{2}{7}x_3 + \frac{1}{28}x_4 - \frac{3}{56}x_5 - \frac{1}{28}x_6 + \frac{3}{56}x_7 = 1250$$

$$\text{代入 } W = 7x_2 + 2x_3 - x_4 + 0.1x_5 - 1.1x_7 - 7000$$

$\Rightarrow W$ 常数项为 0, 原问题有解.

x_1, x_2 为基本可行解.

$$\max z' = -x_1 - x_2 - x_3 = -\frac{2}{7}x_3 - \frac{3}{28}x_4 - \frac{11}{280}x_5 - 2250$$

常数项	x_1	x_2	x_3	x_4	x_5	
z'	-2250	0	0	$-\frac{2}{7}$	$-\frac{3}{28}$	$-\frac{11}{280}$
x_2	1000	0	1	$\frac{2}{7}$	$-\frac{1}{7}$	$\frac{1}{70}$
x_1	1250	1	0	$\frac{2}{7}$	$\frac{1}{28}$	$-\frac{3}{56}$

$$\therefore \max z' = -2250$$

$$\text{即 } \min z = 2250$$

当 $x_1 = 1250, x_2 = 1000, x_3 = 0$ 时取到最小值